

Second Terminal Examination - 2082

Class : XI

Subject: Chemistry (3011)

Full Marks: 75

Time: 3 hrs.

SET - B

Pass Marks: 30

*Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.*

Attempt all the questions.

Group 'A'

[11 × 1 = 11]

Rewrite the best alternative of each question in your answer sheet.

- The oxidation state of N in NH_4NO_3 is
a. -3 & +5 b. +3 & +5 c. +3 & -5 d. -3 & -5
- The mass of ionic species discharged by 1C of electricity is
a. ECE b. CE c. atomic mass d. Molecular mass
- If 0.44 g of a colorless oxide of nitrogen occupies 0.224 l at 2 atm and 273°C, then the compound is
a. NO_2 b. N_2O c. N_2O_2 d. N_2O_5
- When the temperature is raised, the viscosity of the liquid decreases. This is because
a. Decreased volume of solution
b. Increase in temperature increases the average K.E. of molecules which overcomes the attractive force between them
c. Decrease covalent and hydrogen bond formation
d. Increased attraction between molecules
- CO_2 molecule has
a. Three sigma and one pi bond b. Four sigma bonds
c. One sigma and three pi bonds d. Two sigma and two pi bonds
- Which of the following has ionic, covalent and co-ordinate covalent bonds?
a. H_2S b. NaCl c. NH_4Cl d. H_2CO_3
- Formula of functional group for ester is
a. $-\text{CHO}$ b. $-\text{SH}$ c. $-\text{CONH}_2$ d. $-\text{COOR}$
- Cetane rating is used to indicate the quality of:
a. Kerosene b. Diesel c. Gasoline d. LP Gas
- Which of the following is not the test of ozone?
a. test with starch solution b. test with mercury
c. Test with silver d. test with benzidine
- What happens when sulphuric acid is passed through hydrogen sulphide gas?
a. H_2SO_4 is reduced to sulphur b. H_2S is reduced to sulphur
c. H_2SO_4 is reduced to SO_2 d. H_2S is oxidized to SO_2
- Anhydride of sulphurous acid is
a. H_2S b. SO_2 c. SO_3 d. $\text{S}_2\text{O}_3^{2-}$

Group – B

Give short answer to the following questions.

[8 × 5 = 40]

12. The electronic concept of redox reaction is based on electron transfer processes. [1+1+3]
- Define reduction according to the electronic concept.
 - Define disproportionation reaction with example.
 - Balance the following reaction using the oxidation number or ion-electron method.
$$\text{C}_{12}\text{H}_{22}\text{O}_{11} + \text{HNO}_3 \longrightarrow \text{NO}_2 + \text{C}_2\text{H}_2\text{O}_4 + \text{H}_2\text{O}$$

OR

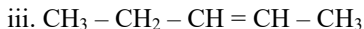
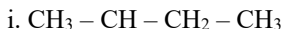
Michael Faraday studied the quantitative aspects of electrolysis by establishing a relationship between the amount of chemical change and the quantity of electricity passed through an electrolyte. [1+1+3]

- State Faraday's second law of electrolysis.
 - Write the relation between the electrochemical equivalent (ECE) and the equivalent weight of a substance.
 - A metallic spoon is coated with silver by passing a current of 5 A through AgNO_3 solution for 5 hours. What is the thickness of silver plating if the area of the spoon is 12 cm^2 ? (density of Ag = 10.5 g/cc and atomic weight of Ag = 108)
13. Using the combined gas equation, the volume of gas liberated at the electrodes during electrolysis can be determined when the temperature and pressure of the system change.
- Derive the combined gas equation. [3]
 - Calculate the pressure required to compress 2000 ml of a gas at 700 mm of pressure and 20°C into a container of 100 ml capacity at temperature of -150°C . [2]
14. Charles' law can be verified by plotting different values of volumes against temperatures. [1+1+1+2]
- State Charles' law.
 - Draw a graph of volume (V) versus temperature ($^\circ\text{C}$) for the given data and extend the graph backward to determine the temperature at which the volume becomes zero.
 - Write the applications of Charles' law.
 - A sample of nitrogen gas occupies 100 cc at 27°C . Calculate the temperature at which its volume will be double, assuming the pressure remains constant.
15. Molecular formulas of some compounds are given below:
- | | |
|--------------------|-----------------------------|
| Beryllium chloride | BeCl_2 |
| Ammonia | NH_3 |
| Nitric acid | HNO_3 |
| Ethene | $\text{CH}_2 = \text{CH}_2$ |
- Predict the geometry of BeCl_2 on the basis of hybridization. [2]
 - Why is ammonia soluble in water? [1]
 - Draw the Lewis structure of i) Nitric acid ii) Ethene [2]

16. The systematic arrangement of organic compounds containing same functional group gives homologous series,
- Mention four characteristics of homologous series. [2]
 - Write the structure of first and second member of alkyne [1]
 - Define catenation property of carbon with examples. [1]
 - A fuel has octane number 85, what does it mean? [1]

OR

- Define functional group. Mention its two significances. [2]
- Write the IUPAC name of following compounds [3]



17. Diatomic oxygen and ozone are allotropes of oxygen.

K_2O	Al_2O_3	NO	CO_2	Fe_3O_4	Na_2O_2
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- Identify the peroxide and mixed oxide from the given table. [1]
- Identify amphoteric oxide and basic oxide with suitable reactions. [2]
- How does CFCs deplete ozone layer? [2]

18. Ammonia is manufactured by Haber's process. Le-Chatelier's principle can be applied in Haber's process for the higher yield of NH_3 .

- What are the necessary physical conditions for the higher yield of NH_3 by Haber's process? [2]
- What happens when NH_3 is passed through CuSO_4 solution till excess? [2]
- Why is NH_3 highly soluble in water? [1]

19. Sulphur belongs to group VIA in the periodic table.

- Discuss the role of H_2S as an analytical reagent with reactions. [2]
- Compare the bleaching properties of SO_2 and Cl_2 . [2]
- What happens when a filter paper dipped in lead acetate is exposed to the gas obtained by treating iron sulphide with dilute sulphuric acid? [1]

Group - C

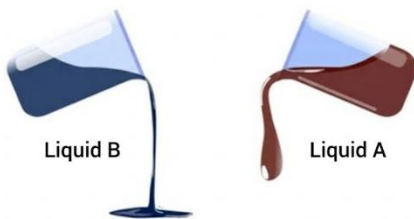
Give long answer to the following questions.

[3 × 8 = 24]

20. There are three distinct physical states of matter, i.e., solid, liquid, and gas.

- Write the statement and derivation of Dalton's law of partial pressure. [2]
- A 1 L sample of dry gas at 25°C has the following compositions: 0.0152 g of Ar and 0.00107 g of CO_2 . What are the partial pressures for each component's gas in the mixture? What is the total pressure? [3]

- c. In the figure below, one beaker contains honey and the other contains water, showing different flow rates. Which property of liquids is responsible for this behaviour, and why do honey and water flow at different rates? Also, identify liquid A and liquid B. [2]



- d. Crystals of washing soda ($\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$) kept in dry air gradually lose their lustre and become powdery. Which phenomenon is responsible for this change? [1]
21. Answer the followings;
- Differentiate between covalent compounds and ionic compounds. [2]
 - What is metallic bond? Why are metals good conductor of electricity? [2]
 - Define dipole moments and mention its applications. [2]
 - On the basis of VSEPR theory, predict the shape of (i) NH_3 (ii) CH_4 . [2]
22. Sulphuric acid is known as the king of chemicals. The industrial use of this acid can be taken as an indicator of the economic development of that country.
- Write down the principle and reactions involved in the manufacture of sulphuric acid by the Contact process. Why is the process of manufacturing sulphuric acid known as the Contact process? [2+1]
 - Draw the flow sheet diagram for the manufacture of sulphuric acid by the Contact process. [2]
 - Write down the reactions which show: [1+1]
 - H_2SO_4 as a dehydrating agent
 - H_2SO_4 as an oxidizing agent
 - Write the molecular formula of hypo and write its one use. [1]
- OR**
- Write down the reactions in which [2]
 - SO_2 acts as reducing agent.
 - NH_3 acts as precipitating agent.
 - What happens when a piece of zinc is added to potassium permanganate solution acidified with dil. H_2SO_4 ? [1]
 - Write the reactions of : [2]
 - conc. HNO_3 with copper
 - conc. HNO_3 with sulphur
 - What is passivity of iron? [1]
 - Differentiate between nascent and atomic hydrogen. [2]

❧ The End ❧